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Problem

In triangle ABC with $CA = CB$, point E lies on the circumcircle of ABC such that $\angle ECB = 90^\circ$. The line through E parallel to CB intersects CA in F and AB in G . Prove that the center of the circumcircle of triangle EGB lies on the circumcircle of triangle ECF .

Solution. Let $\angle BAC = \theta$. Let T be the antipode of C . Observe that $ET \parallel AB$ since $ETBA$ is a rectangle.

Claim 1 — $CEFO$ is cyclic.

Proof. Let (CEF) intersect AT at O' . Then, $\angle EO'C = \angle EFC = \angle AFG = \angle ACB = \pi - 2\theta = 2\left(\frac{\pi}{2} - \theta\right) = 2\angle CBE$. Since $\angle EOC = 2\left(\frac{\pi}{2} - \theta\right) = \angle EO'C$ and $C - O - O'$, this implies that $O' = O$.

Let the perpendicular bisector of EB intersect AB at P . □

Claim 2 — $CEFOP$ is cyclic.

Proof. $\angle BPO = \theta$ and $\angle CPO = \pi - \angle CEO = \pi - \angle CEB = \pi - \angle CAB = \pi - \theta$. □

Claim 3 — P is the circumcenter of $\triangle EGB$.

Proof. Since P lies on the $\perp$ bisector of EB , it is enough to show that $\angle BPE = \angle EGB$.

Since $\angle EGB = \pi - \theta$ and $\angle EPO = \angle ECO = \angle ECT = \frac{\pi}{2} - \angle CTE = \frac{\pi}{2} - \angle CBE = \theta$.

Since $\angle BPO = \theta$, we get that $\angle EPB = 2\theta \implies \angle BPE = 2\pi - 2\theta = 2(\pi - \theta) = 2\angle EGB$ and we are done! □